

MEMS-Based Storage

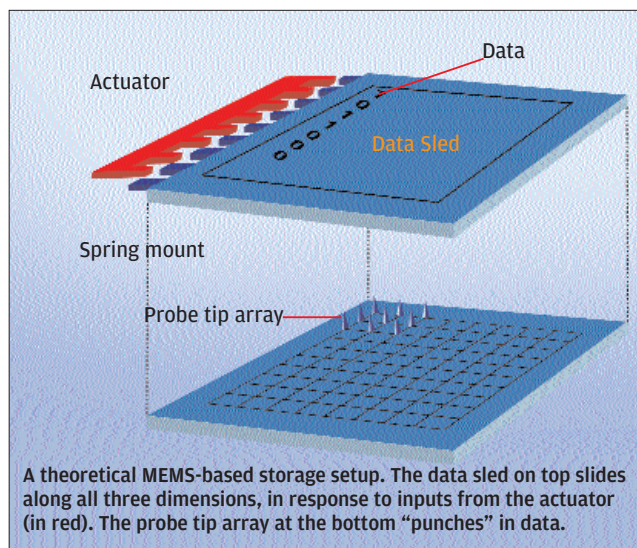
MEMS (Micro-Electro-Mechanical Systems) is, according to memsnet.org, “the integration of mechanical elements, sensors, actuators, and electronics on a common silicon substrate through microfabrication technology.” The mechanical elements referred to here, range in size from a few micrometers to a millimetre. Actuators are just devices that convert an electronic signal to a physical action—for example, the device in the hard disk that is responsible for positioning the head precisely. In fact, we can take the example of a MEMS-based storage system to better explain what MEMS are.

How Does It Work?

Different MEMS storage systems work differently, but we can describe the concept. Take a look at the figure alongside. This isn’t a working system, but just an example of a general MEMS storage system. The data “sled” at the top can move in all three directions; it is spring-mounted over the probe tip array, an array of mechanical tips that do the reading and writing (we aren’t getting into the details here). There’s an actuator on each side of the data sled, and it moves the sled in response to electric currents. Now when the first bit is written, the sled and the tip array are aligned, and then the sled moves along one axis while the tips do their work—writing a 1 or a 0. Note that the sled doesn’t rotate; it slides. Also note that everything in this arrangement is mechanical and electronic.

What’s Being Done?

At CeBit 2005 in Hannover, Germany, IBM showed off a MEMS-based storage device that it said could achieve densities in the range of 1 TB per square inch. The device is called the Millipede, because of the thousands of probe tips. The tips are of silicon, and the data substrate is a material called “plexiglass.” To write a bit of data, a tip is heated to 400 degrees C. When it “pokes” the plexiglass, it softens it and makes a dent there. To read data, the tips are heated to 300 degrees C and pulled across the surface



of the plexiglass. When it falls into a dent, the tip cools down because more surface area comes in contact with the (cooler) plexiglass. The temperature drop reduces its resistance, which can be measured. Finally, to erase a bit, a hot tip is passed over the dent, making it pop back up.

What Promise Does It Hold?

Plenty! MEMS-based storage devices such as Millipede could well be hard disk killers, depending on the research dollars spent. Seek times are lower and more stable than those of hard disks.

In the range of 1 to 10 GB, MEMS-based storage has the lowest cost per byte compared to non-volatile memory and hard disks. Data transfer rates can reach 1 gigabyte per second. Also, MEMS-based devices are smaller and use considerably less power.



Jargon Buster

Spatial Light Modulator (SLM)

An SLM consists of an array of optical elements—pixels—in which each pixel acts independently as an optical “valve” to adjust or modulate light intensity.

Solid Immersion Lens

A type of lens that has a high refractive index, hemispherical in shape, optimised for precision.

Polymers

These are natural or synthetic plastic-like structures where two or more like molecules are joined to form a more complex molecular structure.

Photolithography

This is a process used in semiconductor device fabrication to transfer a pattern from a “photomask” to the surface of a wafer or substrate. Here, a chip, typically of silicon, is coated with a chemical called a photoresist. Flashing a pattern of light and dark onto the photoresist causes it to harden in the areas exposed to light. The parts not exposed to light stay soft and are etched away. (A photomask is a high-precision plate containing microscopic images of electronic circuits.)

Electron beam lithography (EBL)

As opposed to photolithography, in EBL, electron beams are used to create the patterns directly on-chip.

Scanning Tunnelling Microscope

The STM allows scientists to visualise regions of high electron density and hence infer the position of individual atoms.

Atomic Force Microscope (AFM)

This kind of microscope works by scanning a semiconductor tip over a surface. The tip is positioned at the end of a cantilever beam. As the tip is repelled by or attracted to the surface, the beam deflects. The magnitude of the deflection is captured by a laser that reflects at an oblique angle from the very end of the cantilever. A plot of the laser deflection versus tip position on the sample surface provides the resolution of the topography of the surface.

Tunnelling Current

Tunnelling is a quantum mechanical effect. A tunnelling current occurs when electrons move through a barrier that they “shouldn’t” be able to move through. In the quantum mechanical world, electrons have wave-like properties. These waves don’t end abruptly at a wall or barrier, but taper off quite quickly. If the barrier is thin enough, given enough electrons, some will move through and appear on the other side. When an electron moves through the barrier in this fashion, it is called tunnelling.

Cantilever

Simply speaking, a cantilever is a projecting structure supported only at one end, like a diving board.